

[Welcome](#)

[Learning outcomes, workload and...](#)

[Flow Control Theory & PI-Diagrams](#)

✓ [Valve Structure & Sizing](#)

[Teollisuuden instrumentointi/Ind...](#)

[Projektointi](#)

✓ [Laboratoriotyöt](#)

[Indeksi](#)

[Intelligent Field Devices \(FDM\)](#)

[Labra 1](#)



[Valve Structures, Actuators, Positioners](#) ○



[Valmet Neles ND9000 Positioner Manual](#)



[Pump Diagram Sulzer \(140 juoksupyörä\)](#) ○



[Pump Diagram Kolmeks \(105 juoksupyörä\)](#) ○



[Know Pulp \(Pumps\)](#)



[Nelprof Sizing Tool](#)



[PI-kaavio](#) ○



[Venttiilin mitoitusohjelman kysymyspatteri](#)

Mark as done



[Valve Sizing](#)

Due: Friday, 23 October 2026, 8:58 PM

To do ✓

CONTROL VALVE SIZING ASSIGNMENT

Field Devices and Instrumentation

Objective

The objective of this assignment is to learn the fundamentals of control valve sizing by performing a complete sizing exercise independently. During the assignment, students will become familiar with the valve sizing process, the use of sizing software, and the analysis and interpretation of sizing results. The knowledge and skills acquired during the assignment are also essential for the course examination.

Assignment Requirements

1. The assignment is completed **individually**. Each student sizes a control valve for their assigned process application. The process data and operating conditions are provided in the accompanying Excel file.
2. The final outcome is a control valve selected and sized for the process application. The valve selection must be justified using calculations, sizing software results, graphical outputs, and supporting technical documentation.

Report Preparation

The report shall be prepared using the JAMK reporting template and following the JAMK APA 7 referencing guidelines. [\[5a Lähteid...tointiohje | Other Sites\]](#), [\[Other_repo...plate.dotx | Word\]](#)

Please note the following:

1. Use the JAMK reporting template and preserve the original formatting. [\[Other_repo...plate.dotx | Word\]](#)
2. An abstract page is **not required**.
3. A **table of contents** is recommended, even for a short report.
4. **References** shall be presented according to the JAMK APA 7 guidelines. [\[5a Lähteid...tointiohje | Other Sites\]](#)
5. All figures and graphs must be referenced and discussed in the text.

Assignment Procedure

1. Familiarise Yourself with the Process Application

- Study your assigned process application.
- Identify and collect the required sizing data.

2. Review the Pump Curve

- Analyse the pump performance curve.
- Estimate the maximum available pressure.

3. Calculate Pressure Loss Between the Pump and the Valve

- Identify and document all components that cause pressure loss between the pump and the valve.
- Estimate the total pressure loss in the system section under consideration. You may use an online pressure drop calculation tool if appropriate.

4. Calculate the Pressure Drop Across the Valve at the Design Operating Point

- Use the DPM formula described in the *Flow Control Manual*.

5. Perform Valve Sizing

- Enter the process data into the sizing software.
- Perform the initial sizing using an **RE-series segmented ball valve**.

6. Analyse the Sizing Results

- Study the sizing software graphs and results.
- Compare the results with the examples presented in the *Flow Control Manual*.

7. Perform a Comparative Evaluation

- Repeat the sizing exercise using an **M1-series ball valve**.

8. Select the Most Suitable Valve

- Decide whether the RE-series or M1-series valve is more suitable for the application.
- Justify your selection using sizing results, graphs, and technical analysis.

9. Review Product Documentation

- Study the manufacturer's technical documentation for the selected valve.
- Use relevant information from the documentation to support your report.

10. Document All Work

Include the following in your report:

- Process description
- Input data
- Calculations
- Assumptions
- Sizing software outputs
- Graphs and figures (from Nelprof)
- Analysis of results
- Valve selection rationale
- Conclusions

Sources

At a minimum, use the following sources:

- *Flow Control Manual*
- Valve manufacturer documentation (e.g., Valmet)
- Sizing software reports and graphical outputs

All sources shall be cited according to the JAMK APA 7 guidelines. [\[5a Lähteid...tointiohje | Other Sites\]](#)

Assessment

The assignment is assessed on a **Pass / Fail** basis.

To receive a passing grade for the assignment, the student must demonstrate the ability to:

- collect and analyse the required process data,
- perform and document valve sizing calculations,
- use valve sizing software appropriately,
- interpret sizing graphs and results,
- compare alternative valve solutions, and
- justify the final valve selection using technical reasoning.

Completion of the assignment is a mandatory course requirement. The assignment does not directly affect the numerical course grade, but it must be completed successfully before the examination results can be used in the final course assessment.

Assessment Criteria

Assessment Area	Pass	Fail
Report structure and formatting	Report follows JAMK reporting guidelines and the required template is used correctly.	Report structure is incomplete or formatting requirements are not followed.
Process application description	Process application and sizing data are presented clearly and supported with appropriate figures.	Process description or sizing data are missing, unclear, or incorrect.
Calculations and sizing methodology	Pressure losses and DPm calculations are presented and all relevant components are considered.	Calculations are missing, incorrect, or important components are omitted.
Use of sizing software	Both valve alternatives have been sized and the relevant graphs are included and referenced.	Sizing software has not been used appropriately or required results are missing.
Analysis of results	Results are analysed critically and valve alternatives are compared using technical reasoning.	Analysis is missing, inadequate, or technically incorrect.
Use of references	Relevant sources are used and cited according to APA 7 guidelines.	References are missing, incorrect, or inadequately cited.

Conclusions and valve selection	The final valve selection is justified clearly and logically based on calculations and analysis.	Conclusions are missing or the valve selection is not justified.
Use of AI tools	Any use of AI tools supports the student's own learning and analysis. The report demonstrates understanding of the process, calculations, sizing results, and conclusions. The student is able to justify the presented work.	The report appears to be predominantly AI-generated, contains technical inaccuracies or hallucinated content, or does not demonstrate the student's own understanding of the valve sizing process and results.

Note: Failure to address any major assessment area may result in a failing grade. Minor deficiencies do not automatically lead to failure if the overall work demonstrates adequate competence. The assignment is intended to develop practical skills in control valve sizing and to support preparation for the course examination.

Use of Artificial Intelligence

Generative AI tools may be used to support writing, language checking, and information retrieval. Any use of AI must be **documented in the report**, including a brief description of how the tool was used and for which tasks. The student remains fully responsible for the technical accuracy of all calculations, analysis, results, and conclusions. Control valve sizing is a technical engineering task in which AI tools may generate incorrect, misleading, or hallucinated information. The submitted work must clearly demonstrate the student's own understanding of the valve sizing process and the interpretation of the results. If the report appears to be predominantly AI-generated, contains unverified AI-generated content, or fails to demonstrate the student's own learning and engineering judgement, the assignment may be rejected. The use of AI must support, not replace, the student's own analysis and learning.

Submission

- 1. Submit the report to Moodle **before the exam**.
 - 2. Completing the assignment will significantly support your preparation for the examination.
- Sources: 5a Lähteiden käyttö (APA7-ohje, syksystä 2024 opintonsa aloittaneet) - Opinnäytetyön raportointiohje, [Other reporting template.dotx](#) [\[5a Lähteid...tointiohje | Other Sites\]](#) [\[Other repo...plate.dotx | Word\]](#)

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Additional Material

- [Lessons in Industrial Instrumentation : Tony R. Kuphaldt : Free Download, Borrow, and Streaming : Internet Archive](#)
- [Virtaustekniikka : vedensiirtojärjestelmien toiminnallinen suunnittelu nykyaikaisin menetelmin | Jyväskylän ammattikorkeakoulu | Janet Finna](#)
- [Flow technology.: modern functional design of water transmission systems | Jyväskylän ammattikorkeakoulu | Janet Finna](#)
- [Säätöventtiilidiagnostiikan analysointi ja visualisointi pilvipalvelussa : Control valve PARAMeter ANALyzer 2.0 - Theseus](#)

 [Labratory 3d model](#)

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- Suomi (fi)
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- Русский (ru)
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